

Aspect (metric)	3DGS (baseline)	Mip-Splatting (CVPR'24)	FlashGS (CVPR'25)	Speedy-Splat (CVPR'25)	GSCore (ASPLOS'24)	Reduced-3DGS (2024)	4D-GS (CVPR'24)
Primary goal	Real-time high-quality radiance-field rendering	Anti-aliasing / scale-robust quality	Large-scale + high-res rendering speed	Speed via sparse pixels/primitives + pruning	Hardware/arch acceleration (mobile GPU)	Cut memory footprint (disk/model)	Real-time dynamic scenes
Reported real-time FPS	≥ 30 FPS @ 1080p (GitHub)	(Not the main claim; focuses on quality) (CVF Open Access)	≥ 125.9 FPS (minimum), up to 4K+ scenes (CVF Open Access)	Reports 6.71× rendering speedup (avg) (CVF Open Access)	Arch eval: 15.86× speedup vs mobile consumer GPU (Jaewoong Sim)	1.7× rendering speedup (arXiv)	82 FPS @ 800×800 (RTX 3090) (CVF Open Access)
Speedup vs 3DGS	1×	~quality-focused (no speedup claim) (CVF Open Access)	7.2× avg speedup (CVF Open Access)	6.71× avg speedup (CVF Open Access)	15.86× avg (mobile GPU baseline) (Jaewoong Sim)	1.7× (arXiv)	(Different problem: dynamic; not directly comparable) (CVF Open Access)
Quality on Mip-NeRF 360 (PSNR/SSIM/LPIPS)	27.21 / 0.815 / 0.214 (CVF Open Access)	27.79 / 0.827 / 0.203 (same-scale test) (CVF Open Access)	(Quality preserved while speeding up; numbers depend on eval setup) (CVF Open Access)	“Maintain high image quality” across Mip-NeRF360/T&T/DB (CVF Open Access)	(Arch paper emphasizes perf/energy; quality checked via functional outputs) (Jaewoong Sim)	“Negligible loss” while shrinking size (paper reports standard datasets) (arXiv)	“Comparable or better quality” vs prior dynamic SOTA (CVF Open Access)
Multi-scale / anti-aliasing quality	Can alias when scaling	Strong gains in multi-scale testing (Table 3) (CVF Open Access)	Not main focus	Not main focus	Not main focus	Not main focus	Not main focus
Model size / storage reduction	Large (often hundreds MB–GB)	Similar order to 3DGS	Not main focus	10.6× smaller model (avg) (CVF Open Access)	Hardware design (area/energy focus)	27× size reduction on disk (arXiv)	Dynamic representation includes 4D components (size varies) (CVF Open Access)
Training speed	Baseline (e.g., 30k)	Similar order	Not main focus	1.47× faster	N/A (arch eval)	Not primarily	Training time claims

	iters typical)			training (avg) (CVF Open Access)		training speed	vary; repo notes optimizations in some implementations (CVF Open Access)
Scales to city / very high resolution	Works, but heavy	Quality method	Explicitly targets city-scale up to 2.7 km² + 4K (CVF Open Access)	Helps via fewer Gaussians/intersections	Mobile-oriented throughput/energy	Storage helps deployment	Dynamic scenes; not “city-scale” claim
Mobile/energy emphasis	Not primary	Not primary	Not primary	Useful via pruning	Yes: speed/area/energy analysis (Jaewoong Sim)	Helps mobile download/storage	Not primary
Key technique category	Tile-based raster + visibility + densification	Anti-aliasing via mip/filtered splatting	Kernel + algorithmic optimizations + load balance/pipelining (CVF Open Access)	Sparse pixels + tight intersections + pruning pipeline (CVF Open Access)	Dedicated architectural support for GS pipeline (Jaewoong Sim)	Pruning + adaptive SH + quantization/codebook (arXiv)	3D Gaussians + 4D neural voxels + deformation prediction (CVF Open Access)